

SELECTIVE LONG RANGE DISPERSION FORCES BETWEEN LARGE SYSTEMS

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Two large molecular systems capable of giant dipole vibrations of frequencies ω_1 and ω_2 respectively may exert selective very strong long range forces upon each. The selectivity refers to frequency resonance ($\omega_1 \approx \omega_2$). Further conditions are (i) the dielectric constant of the medium between them shows strong dispersion in the relevant frequency region or (ii) coherent vibrations are excited by pumping. The latter is always attractive, the former may be either attractive or repulsive.

The interaction between two large systems (e.g. giant molecules) at a distance R , larger than their linear dimensions L arises principally from the interaction between their giant dipole vibrations (= zero wave number polarisation waves). Let ω_1 and ω_2 respectively ($\omega_1 \geq \omega_2$) be the frequencies of these vibrations in the separated systems. Then through their interaction at finite distance two frequencies ω_+ , ω_- of the joint system arise,

$$\omega_{\pm}^2 = \frac{1}{2} (\omega_1^2 + \omega_2^2) \pm (\Delta^2 + \beta^4/\epsilon_{\pm}^2)^{\frac{1}{2}},$$

$$\Delta = \frac{1}{2} (\omega_1^2 - \omega_2^2) \quad (1)$$

where $\epsilon_{\pm}$ is the longitudinal dielectric constant at frequency $\omega_{\pm}$ of the medium separating the systems and $\beta^2 = e^2 |A| (z_1 z_2)^{\frac{1}{2}} / MR^3$.

Here the two systems are considered to contain z_1 and z_2 elastically bound particles of mass M and charge e . $|A|$ is a positive constant of magnitude unity which depends on various angles. The frequencies ω_1 and ω_2 depend on the macroscopic shape of the systems and for simplicity will be assumed non degenerate. Generalisations to more complicated systems are straight forward. Clearly the frequency separation increases with the size of the systems and for large systems may reach considerable values such that in a dispersion region of the intermediate medium ϵ_+ may differ considerably from ϵ_- . The result is a resonance-like interaction energy between the systems which is linear in β^2 (i.e. $\propto 1/R^3$) if $\epsilon_+ \neq \epsilon_-$ but $\omega_1 \approx \omega_2$. For $\epsilon_+ = \epsilon_-$ on the other hand the second order London interaction ($\propto 1/R^6$) is obtained

which is not frequency selective.

The interaction energy I is defined as the difference of the free energy F at distance R from its value at $R \rightarrow \infty$. Thus in the extreme quantum case (I_q) when $kT/\hbar$ is small compared with ω_- ,

$$I_q/kT = \hbar ((\omega_+ + \omega_-) - (\omega_1 + \omega_2))/2kT \quad (3)$$

whereas in the extreme classical case

$$I_c/kT = \log \omega_+ \omega_- / \omega_1 \omega_2. \quad (4)$$

To discuss the result consider first the case of weak coupling defined as

$$(\Delta^2 + \beta^4/\epsilon_{\pm}^2)^{\frac{1}{2}} - \Delta \ll \omega_2^2 \quad (5)$$

Systematic development near resonance then yields

$$I_q/kT = \frac{\hbar}{4kT} \left\{ \Delta \left(\frac{1}{\omega_2} - \frac{1}{\omega_1} \right) + \left(\frac{\Delta^2}{\omega_1^2} + \frac{\beta^4}{\omega_1^2 \epsilon_+^2} \right)^{\frac{1}{2}} - \left(\frac{\Delta^2}{\omega_2^2} + \frac{\beta^4}{\omega_2^2 \epsilon_-^2} \right)^{\frac{1}{2}} \right\} \quad (6)$$

Hence at resonance, $\omega_1 = \omega_2 = \omega$, $\Delta = 0$,

$$\frac{I_{q \text{ res}}}{kT} = \frac{\hbar \omega}{4kT} \frac{\beta^2}{\omega^2} \left(\frac{1}{\epsilon_+} - \frac{1}{\epsilon_-} \right), \quad \omega_1 = \omega_2. \quad (7)$$

If $\hbar \omega / 4kT$ is large, $I_q/kT > 1$ may well arise although (5) requires $\beta^2/\omega^2 \epsilon < 1$. The result in the classical case is similar to (7) except for the factor $\hbar \omega / kT$ so that $I_c < kT$. Outside the resonance region the usual London result $\propto 1/R^6$ obtains; eq. (6) must then be taken to the next order.

If the weak coupling condition (5) is not fulfilled

then according to (1) considerable softening of the mode ω_- may arise. The case $\omega_-^2 < 0$ is, of course, well known from the theory of mode softening in ferroelectrics and will not be considered here. Small but positive ω_-^2 implies that the classical limit only need be considered. According to (4) it may lead to strong attractive interaction but is not sensitive to resonance conditions.

Our main result thus indicates the possibility of strong, frequency selective, long range forces between large systems provided the longitudinal dielectric constant $\epsilon(\omega)$ of the intermediate medium shows appropriate dispersion in the relevant frequency region; the sign of $\partial\epsilon/\partial\omega$ decides whether these forces are attractive or repulsive.

Selective long range forces which are always attractive are obtained if the system is pumped as has been suggested to occur in biological systems through their metabolism [1]. It has been shown then that the lowest frequency in a band of polarisation waves should be very strongly excited. Now at infinite separation of our two systems the frequency of the polarisation waves with zero wave number (responsible for long range interaction) is not in general the lowest one but according to (1), ω_- is depressed at closer approach and this may become the lowest frequency at a

distance below a critical one. The interaction then is always negative, (n_- = number of quanta in ω_-)

$$I = (n_- + \frac{1}{2}) \hbar(\omega_- - \omega_2) \approx -\hbar(n_- + \frac{1}{2}) ((\Delta^2 + \beta^4/\epsilon^2)^{\frac{1}{2}} - \Delta)/2\omega_2 \quad (8)$$

Thus we have found that a system capable of giant dipole vibrations may attract another strongly, and selectively in frequency. The range of this interaction is much larger than that of the more usual forces. At closer distance, however, the behavior of the individual atoms composing the large systems should become more relevant and the highly specific nature of the London forces as demonstrated by Jehle et al. [2] as well as other chemical interaction may become of principal importance. The two aspects are complementary and should be decisive in many aspects of selective and specific interaction of large systems [3,4].

References

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